

Edrill Bilan

Manila, Philippines | efbilan@gmail.com | 0956 441 6623 | [efbilan.vercel.app](https://edrill.vercel.app)

linkedin.com/in/edrill-bilan/ | github.com/PyroJayxX

Education

Pamantasan ng Lungsod ng Maynila BS in Computer Science <i>Tertiary Education</i> – Magna Cum Laude, GWA: 1.35 (Graduation: Aug. 2026)	Sept 2022 – Present
Universidad de Manila STEM Strand <i>Senior High School</i> – Graduated with High Honors, GWA: 95.0	Oct 2020 – July 2022

Skills

Programming Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML, CSS, x86 Assembly
Machine Learning: PyTorch, TensorFlow, ChromaDB, RAG, scikit-learn, Pandas, NumPy, Matplotlib, SciPy
Backend Development: MySQL, PostgreSQL, Django, Flask, ExpressJS, FastAPI, Firebase, Supabase
Frontend & Other Tools: ReactJS, NextJS, Tailwind CSS, JavaFX, Git, Figma, Jupyter Notebook, Ollama

Experience

Crawlers Insurance Agency, Inc. AI Intern · Remote, Quezon City, Philippines	June 2026 – Present
Developed CrawlLearn, an insurance edtech platform featuring an AI tutor powered by a RAG pipeline that chunks and embeds course transcripts using Gemini embedding-001, retrieves via pgvector similarity search with Cohere Rerank, and generates on-the-spot quizzes, grounded Q&A, and AI-graded real-world scenarios.	
National Cheng Kung University Research Intern · Computer Vision and Multimedia Laboratory, Tainan, Taiwan	Oct 2025 – Nov 2025
International research collaboration with NCKU's Computer Vision and Multimedia Laboratory, finetuning ResNet-50 w/ multi-scale transformers for Pulmonary Lung Nodule Detection on Chest X-ray images as part of undergraduate thesis; published on MDPI's Q1 Journal of Diagnostics.	
Goldilocks Bakeshop, Inc. IT Service Management Intern · Greenfield Building, Mandaluyong City	June 2025 – July 2025
Assisted in maintaining IT infrastructure while coordinating service requests such as technical support for hardware & software issues, streamlining issue resolution workflows across departments.	

Projects

(Machine Learning) OmniJuris – PHL Legal Intelligence RAG System (2026)	
- Developed a RAG pipeline over 3.2M chunks of the Philippine OmniCorpus with multilingual BGE-M3 embeddings, ChromaDB vector search, cross-encoder reranking, and a locally-hosted Qwen3 4B LLM with SQLite-backed conversational memory context-aware legal queries about Philippine Law. - Stack: Python, FastAPI, React, TypeScript, ChromaDB, sentence-transformers, Ollama, SQLite	
(Machine Learning) Atlas990 – USA Nonprofit Organizations Intelligence System (2026)	
- Developed a full-stack nonprofit lead intelligence platform over IRS Form 990 public tax data, featuring an XGBoost-based priority scorer (0-100) and FAISS-powered vector similarity search for lookalike orgs. - Engineered features from raw IRS Form 990 XML filings public government data, cleaning and transforming public tax records into an optimized Parquet dataset for model training. - Stack: Python, FastAPI, React, TypeScript, Pandas, NumPy, scikit-learn, Parquet, XGBoost, FAISS	
(Deep Learning) Pulmonary Lung Nodule Detection System (2026)	
- Architected RNet-MST, a ResNet-50 enhanced with stage-wise Multi-Scale Transformer blocks and a spatial attention module for attention-based localization – improving recall from $88.02 \pm 1.92\%$ to $91.55 \pm 1.41\%$, with a 12.3% sensitivity gain on small nodules sized at $<70 \times 70$ pixels on Chest X-Ray Images. - Stack: Python, PyTorch, OpenCV, Pandas, NumPy, Jupyter Notebook	

(Web Crawler) TrabaHound (2026)

- Developed a full-stack job aggregation platform that crawls the web for job listings, parses resumes via PDF extraction, and utilizes LLM to identify skill gaps and recommend matching roles.
- Stack: NextJS, TypeScript, Python, Jobspy, LLMs, PDF Parsing

(Full-Stack) MAFA-Inventi (2025)

- Co-developer for a property management system developed for the Inventi Hackathon enabling property managers to upload floor plans, pin issues, and efficiently manage properties, tenants, and maintenance.
- Implemented an LLM-powered dashboard for property analytics and insights.
- Stack: ReactJS, Tailwind CSS, NodeJS, ExpressJS, Supabase, Vercel

(Full-Stack) DOM Compiler (2024)

- Architected and developed a programming language from scratch, building an end-to-end compiler pipeline from lexical to semantic analysis, and syntax parsing with customized keywords, syntax rules, and logic.
- Stack: NextJS, Python, Tailwind CSS, Flask, React, NodeJS

Extracurricular

• Data Engineering Pilipinas | DataCamp Scholar

DataCamp x Data Engineering Pilipinas Partnership Scholar 2026

• Inventi Asia Hackathon | Creativity Award Winner

Smart Property Solutions Hackathon 2025

• Google Developer Student Club (GDSC) Volunteer | Web Development Team

October 2024 – October 2025

• Google Developer Student Club (GDSC) Volunteer | Design and Branding Team

November 2023 – October 2024

• DSPC Annual Journalism Competition | 3rd Place Winner in Online Publishing (English)

Manila Division Schools Press Conference 2019

Certifications

- Intermediate Machine Learning – **Kaggle** (2026)
- Supervised Learning with scikit-learn – **DataCamp** (2026)
- GitHub Foundations Track – **DataCamp** (2026)
- Network Support and Security Course – **Cisco** (2025)

Publications

Bilan, E. F., Manduriaga, E. T., et al. (2026). RNNet-MST: A ResNet-50 with Multi-Scale Transformer Blocks for Pulmonary Nodule Classification and Attention-Based Localization on Chest X-Ray Images. *Diagnostics*, 16(10), 1574. DOI: 10.3390/diagnostics16101574

- Undergraduate thesis developed through a PLM–NCKU international research collaboration with the Computer Vision and Multimedia Laboratory at National Cheng Kung University, Taiwan.
- Published in MDPI Diagnostics, a Q1 peer-reviewed journal, indexed in Web of Science and Scopus.

References

Dr. Hernando Salapare III, *Researcher at CNRS - IS2M, France*
Thesis Adviser, University Research Center PLM

+33 6 52 28 66 07
hssalapare@up.edu.ph

Bren Alfred España, *Business Solutions Manager, JTI*
Professor, Pamantasan ng Lungsod ng Maynila

+63 942 661 1372
bajespana@plm.edu.ph